D1.1 DNA replication

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-d1-1.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 604–614 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

How DNA is copied

D1.1.6 DNA polymerases work 5′ to 3′

pp. 608–609

DNA polymerase joins the 5′ phosphate of an incoming nucleotide to the free 3′ -OH at the end of the growing strand, using energy from the nucleotide's extra phosphates. So new strands grow only **5′ to 3′**, and the polymerase needs an existing 3′ end, a **primer**, to start.

D1.1.7 Leading and lagging strands

pp. 609–611



Because the strands are antiparallel, one new strand, the **leading strand**, is made **continuously** towards the fork from a single primer. The other, the **lagging strand**, is made **discontinuously** away from the fork as short **Okazaki fragments**, each needing its own primer.

D1.1.8 The enzymes

pp. 610–612

In prokaryotes, helicase separates the strands; **DNA primase** makes short RNA primers; **DNA polymerase III** extends them with DNA nucleotides; **DNA polymerase I** removes the RNA primers and replaces them with DNA; and **DNA ligase** joins the fragments into a continuous strand.

D1.1.9 Proofreading

pp. 613–614

If DNA polymerase III adds a nucleotide with a mismatched base, it removes it from the 3′ end and adds the correct one. Proofreading cuts the error rate about a hundredfold, keeping mutations rare. Taq polymerase, used in PCR, cannot proofread.

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